Network Connectivity and Troubleshooting Tools

Question 1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Answer: The **ping** command checks whether a destination can be reached over an IP network. It sends ICMP Echo Request packets and waits for Echo Reply packets. The reply time is usually shown in milliseconds (ms).

Step-by-step procedure (Windows):

1. Connect the computer to the internet.
2. Press **Windows + R**, type **cmd**, and press Enter.
3. Type the command below and press Enter:

```
ping www.youtube.com
```

4. Wait until the command finishes. Windows normally sends four requests by default.
5. At the bottom, find the line beginning with **Minimum, Maximum, Average**. Record the value after **Average** in milliseconds.

Example output (illustrative only):

```
Pinging youtube.com [142.250.XXXX.XX] with 32 bytes of data:
Reply from 142.250.XXXX.XX: bytes=32 time=24ms TTL=117
Reply from 142.250.XXXX.XX: bytes=32 time=27ms TTL=117
Reply from 142.250.XXXX.XX: bytes=32 time=23ms TTL=117
Reply from 142.250.XXXX.XX: bytes=32 time=26ms TTL=117

Ping statistics:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
Approximate round trip times in milli-seconds:
    Minimum = 23ms, Maximum = 27ms, Average = 25ms
```

> **Result to record:** Average response time = **25 ms in this example**. This is a sample value, not a measurement from your computer. Run the command on your own computer and replace it with the actual Average value shown.

Interpretation: A lower response time generally means a quicker round trip between your computer and the destination. Results can vary with Wi-Fi strength, internet traffic, routing, and the destination server. Some websites or networks block ICMP, so a ping timeout alone does not prove the website is down.

Question 2. Run traceroute (tracert on Windows) to www.flipkart.com and list all intermediate hops shown in the output.

Answer: The Windows command **tracert** displays the network hops that replies are received from along the route toward a destination. Each numbered line is a hop, often a router or other Layer 3 network device. The route can change depending on the network and time.

Step-by-step procedure:

1. Open Command Prompt (Windows + R → type **cmd** → Enter).
2. Run the following command:

```
tracert www.flipkart.com
```

3. Wait for the trace to finish. It can take a while if some hops do not respond.
4. Read the output from top to bottom. The numbers 1, 2, 3, and so on identify the hop order. Copy each hop's IP address or host name exactly as displayed.
5. If a line displays *** * *** or “Request timed out,” record that hop as “No response” rather than guessing its address.

How to present your hop list:

```
Example format only – not actual Flipkart test results:
Hop 1: 192.168.1.1
Hop 2: 10.XX.XX.1
Hop 3: 100.XX.XX.1
Hop 4: * * * (no response)
Hop 5: [IP address or host name shown]
...
Final hop: [destination address, if reached]
```

> **Result:** The intermediate hops are the numbered entries shown by your own **tracert www.flipkart.com** output. Replace the example entries above with every line from your test, in order. I cannot truthfully list your actual hops without seeing the output from your computer.

Important notes: Private addresses (such as 192.168.x.x, 10.x.x.x, and 172.16.x.x–172.31.x.x) may appear inside local or provider networks. Asterisks do not necessarily mean the route is broken; routers may be configured not to answer traceroute probes. The final destination may also block probes.

Question 3. Use nslookup to find the IP address of www.zomato.com and record the result.

Answer: nslookup queries the Domain Name System (DNS) to find the records associated with a domain name. An A record returns an IPv4 address; an AAAA record returns an IPv6 address. A website may have several addresses, and the results can vary by DNS resolver, location, and time.

Step-by-step procedure:

1. Open Command Prompt.
2. Type the command and press Enter:

```
nslookup www.zomato.com
```

3. Find the “Name” and “Address” entries in the answer section. The first “Server” and “Address” lines usually identify the DNS resolver used for the query; do not confuse that resolver address with Zomato's address.
4. Record every returned address belonging to **www.zomato.com**. If you need IPv4 only, use:

```
nslookup -type=A www.zomato.com
```

Example format (illustrative only):

```
Server:  [your DNS resolver]
Address: [resolver IP]

Non-authoritative answer:
Name:    www.zomato.com
Addresses: [IP address returned]
          [another IP, if returned]
```

> **Result to record:** Write the exact IP address or addresses returned on your own computer. The placeholders above are intentional; DNS answers can change and should not be invented or treated as live test results.

Interpretation: If nslookup returns an address, DNS resolution worked for that query. If it reports a timeout, “Non-existent domain,” or another error, check the spelling, internet connection, and DNS resolver. A DNS result does not by itself prove that the website's web server is reachable.

Question 4. Test connectivity to another device on your local network using ping and describe what happens if the device is offline.

Answer: To test a phone or another computer on the same Wi-Fi/LAN, find its local IP address and ping that address from your computer. Both devices should normally be connected to the same local network, and the target device must permit ping replies.

Step-by-step procedure:

1. On the target device, find its IP address. On Windows, run **ipconfig** and look for the active adapter's IPv4 Address. On a phone, check the connected Wi-Fi network details.
2. On your computer, open Command Prompt.
3. Replace the example address below with the target device's actual local IP:

```
ping 192.168.1.20
```

4. Read the replies and packet statistics. Successful replies show response times; the summary reports packets sent, received, and lost.

If the device is offline: The ping may show “Request timed out” or “Destination host unreachable,” and the summary may show packet loss (possibly 100%). This means no ping reply was received; it does not prove the device is offline in every case. The device may be asleep, on a different network, blocking ICMP, or isolated by the router's Wi-Fi settings.

Example observations:

```
Device online and responding:
Reply from 192.168.1.20: bytes=32 time=3ms TTL=64

Device offline or not replying:
Request timed out.
Request timed out.
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

Conclusion: If replies are received, there is IP-level connectivity to the target for ping. If no replies are received, confirm the IP address, Wi-Fi/LAN connection, and target firewall/settings before concluding that the device is offline.

Question 5. Write a 5–7 line report explaining which tool is most useful for website not loading, slow connection, and suspected DNS issues. Give examples based on the tests above.

Answer — short report (7 lines):

1. **Ping** is useful for checking basic reachability and round-trip delay; in Question 1, ping www.youtube.com and record the actual average time shown by your computer.

2. If YouTube ping replies are received with low and consistent times, the route to that host is responding, although this does not guarantee that the website itself will load.

3. For a slow connection, **ping** can reveal high response times or packet loss; compare several replies and, if possible, test more than one destination.

4. **Tracert** helps investigate a route problem; in Question 2, tracert www.flipkart.com shows the responding hops and any hops that do not reply.

5. A hop showing asterisks may filter probes, so it is not enough by itself to identify the exact fault.

6. **Nslookup** is most useful for suspected DNS issues; in Question 3, nslookup www.zomato.com shows whether the domain resolves to an IP address.

7. For a local-device check, Question 4's ping test can show replies or packet loss, but a timeout can also be caused by firewall or Wi-Fi isolation settings.

Quick command reference

```
ping www.youtube.com
tracert www.flipkart.com
nslookup www.zomato.com
nslookup -type=A www.zomato.com
ping <target-device-local-IP>
```

Submission reminder: This document provides the procedures and clearly marked examples. To make the report reflect your own test results, run the commands and replace the sample/placeholder values with the exact output from your computer.